

Quant Problem Solutions

A working notebook of problems and solutions

Cosmo Wu

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Preface

This document is a personal collection of quantitative problems I have worked through, along with my solutions. The goals of this notebook are:

- To force myself to write up clean, end-to-end solutions rather than scratch work.
- To tag and categorize problems so I can revisit weak areas.

Each problem is presented with a difficulty rating, tags, topics, the problem statement, and my full solution. Note that the problems are not in any particular order and are just in the order of when I solved them. If I'm for some reason not allowed to show certain problems here, please contact me at cosmoyangwu@gmail.com. Enjoy!

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MEDIUM

Tags: brainteasers

Topics: Events

Date: May 4, 2026

1. Repeated Flipper

Problem

25 fair coins are lined up and flipped once. Afterwards, remove all of the coins that showed tails, and repeat the aforementioned process until you have no coins left or all of the coins show up heads. Find the probability that you end the game by having all of the coins show heads.

Solution. Note that for the game to end, you must either have all heads or all tails at the final step. There is intrinsic symmetry, so our answer is simply $\boxed{\frac{1}{2}}$.

MEDIUM

Tags: Brainteasers

Topics: Other

Date: May 4, 2026

2. Sum1

Problem

Find two real numbers x and y composed of only 1s in their decimal expansion such that $xy = x + y$. What is xy ?

Solution. To solve, first note we can rewrite this as $x = \frac{y}{y-1}$. We must then use the property of x first to note that our expression must be an expression of 1s. We use this fact to know that $y = 1.1$ because otherwise, we wouldn't get all 1s for x . Thus, $x = 11$ and so $xy = 12.1$. The key is to isolate effects to utilize known properties.

MEDIUM

Tags: Brainteasers

Topics: Other

Date: May 4, 2026

3. Mathematical Birthday

Problem

Your grandpa from the 19th and 20th centuries stated that he was $x^2 + y^2$ years old in the year $x^4 + y^4$, $2m$ years old in the year $2m^2$, and $3n$ years old in the year $3n^4$. All the variables are integers. Determine the value of $xymn$.

Solution. First, to get $x^4 + y^4$ between 1900 and 2000, we must have $x = 5$ and $y = 6$ (the other way around works too, since we have symmetric equations! :D). Then, we have $625 + 1296 = 1921$, and so our grandpa is 61 years old, meaning he was born in 1860. We then must have $1860 + 2m = 2m^2$, which gives us $m = 31$. Similarly, $1860 + 3n = 3n^4$, which yields $n = 5$. Thus, our answer is $5 \times 5 \times 6 \times 31 = \boxed{4650}$.

MEDIUM

Tags: Brainteasers

Topics: Combinatorics

Date: May 4, 2026

4. Picture Day

Problem

Ten students of distinct heights are lining up for a picture. The photographer requires that the two tallest students stand in the two center positions and that the remaining students line up such that the heights strictly descend outwards. How many ways can the students line up?

Solution. This is a classic constructive counting problem! First, for our center students, we have 2 ways to line them up. Next, for each of the remaining 8 students, consider choosing 4 of them for the left side. Since the height must be in correct order, how the 4 students for each side are ordered is already determined. Thus, our answer is $2 \cdot \binom{8}{4} = \boxed{140}$.

MEDIUM

Tags: Probability

Topics: Combinatorics

Date: May 4, 2026

5. Ace Distribution

Problem

Assume the dealer holds a fair deck of 52 cards. She shuffles the deck and gives each of you and your three friends at the table 13 cards. What is the probability that each of you receive an ace?

Solution. Suppose the first card for each of us is an ace. Then, for the remaining cards, we can split them up in $\frac{48!}{12!^4}$ ways. This is simply given from splitting up the 48 cards into each of the friends and then accounting for the fact that order doesn't matter within one person's cards. The total number of ways is then $\frac{52!}{13!^4}$. We have $4! = 24$ ways of how each of us gets an ace (like what suit and what not). Thus, our answer is

$$24 \cdot \frac{48! \cdot 13!^4}{52! \cdot 12!^4} = \boxed{\frac{2197}{20825}}$$

MEDIUM

Tags: Brainteasers

Topics: Other

Date: May 4, 2026

6. Say 50!

Problem

You and your friend are playing a game where you each pick an integer from 1 to 10 inclusive and add that to a running total. The person that states 50 is the winner. You get to go first, what number do you say first?

Solution. The key to solving this problem is to realize that we can start from the bottom up. If the game is to just say 10, then if you start first, you're guaranteed to win. If you're starting at 11, then you're guaranteed to lose. This includes all multiples of 11. Meaning, if you start at 44, you will lose. Thus, if we say $\boxed{6}$, my friend will be essentially starting at 44 and so will lose!

EASY

Tags: Brainteasers

Topics: Combinatorics

Date: May 5, 2026

7. 2D Paths I

Problem

You are playing a 2D game where you are trapped within a 6×6 grid and navigate via points on the grid. You start at the bottom left: point $(0,0)$. You can only move up and right. How many possible paths can you take to get to the top right point: $(6,6)$?

Solution. First, note that for each path, there is a 1 – 1 correspondence to how you arrange 6 letters of R and 6 letters of U . Thus, our answer is $\binom{12}{6} = \boxed{924}$.

EASY

Tags: Probability

Topics: Discrete Random Variables, Expected Value

Date: May 5, 2026

8. Coin Runs

Problem

You toss a fair coin 100 times and record the outcomes. How many runs will you have on average? A run is classified as the longest continuous flips of heads or tails.

Solution. I sadly got this question wrong when solving it. However, the solution is to consider adjacent coins X_i and X_{i+1} . Between 2 coins, the probability that X_{i+1} is a new coin is $\frac{1}{2}$. Thus, for $X_2, X_3, \dots, X_{100}$, the probability that each is a new run is $\frac{1}{2}$, and for X_1 , the probability is 1. Thus, by summation of expectation, the number of runs on average would be $1 + \frac{99}{2} = \boxed{50.5}$.

EASY

Tags: Brainteasers

Topics: Other

Date: May 5, 2026

9. Revolver Time

Problem

Jenny is firing her revolver, which has 6 chambers. The time from the first shot to the last is 1 minute. Assuming a uniform firing rate, how many seconds will it take Jenny to fire 3 shots?

Solution. We have 5 gaps, meaning each turn takes $\frac{60}{5} = 12$ seconds, and so it would take $\boxed{24}$ seconds.

MEDIUM

Tags: Brainteasers

Topics: Games

Date: May 6, 2026

10. Greedy Pirates

Problem

A treasure of 100 gold coins must be divided among five pirates, ranked by seniority from 1 (least senior) to 5 (most senior). They follow these rules to split the coins:

1. The most senior pirate proposes how many coins to distribute to each pirate.
2. All pirates vote on the proposal, including the pirate that made the proposal. If 50% or more of the pirates accept the proposal, the coins are distributed as proposed.
3. Otherwise, if more than 50% of the pirates reject the proposal, the most senior pirate is tossed overboard. The process restarts with the next most senior pirate making a proposal and the remaining pirates casting votes.
4. This process repeats until a proposal is accepted or only one pirate remains, in which case the last pirate gets all 100 gold coins.

Pirates prioritize survival over wealth. The pirates will act rationally to maximize their gain while ensuring they survive. How many coins does the most senior pirate (pirate 5) get if he makes an optimal proposal? If the most senior pirate is always thrown overboard, answer 0.

Solution. This is another classic "induction" question where we must consider the case of 1 pirate, then 2 pirates, and etc. For 1 and 2, the most senior pirate can always get all the coins. If we have 3 people, then we would want the most senior to give person 1 1 coin so that he doesn't throw person 3 (most senior) out. This is the case since if the most senior does get thrown out, person 1 will end up getting 0 (person 2 will take all of it). Similarly, if we have 4 people, the most senior would want to give a coin to person 2. With this logic, person 5 would give a coin to both 1 and 3. So our answer is 98.

MEDIUM

Tags: Statistics

Topics: Events

Date: May 6, 2026

11. German Tanks**Problem**

Suppose that German tanks are assigned distinct serial numbers $1, 2, \dots, N$. You observe 6 tanks with serial numbers 38, 17, 59, 42, 97, and 120. Under a frequentist approach, what is the best guess for N ?

Solution. I also was unable to solve this problem successfully :(However, the key to this problem is to notice that we can reframe this problem as an ace problem where the 120 serves as one of the checkpoints. Since we chose 5 other numbers before 120, that means we have $114/6 = 19$ as our gap. Thus, our closest value is $N = 120 + 19 =$ 139.

MEDIUM

Tags: Probability

Topics: Combinatorics, Conditional Probability

Date: May 6, 2026

12. Consecutive Tails

Problem

A coin is flipped 10 times and the outcomes are recorded. Find the probability that any tails occur only in consecutive pairs. For example, with 4 flips, $TTHH$, $TTTT$, and $HHHH$ are both valid, but $HTHH$ and $TTTH$ are not valid.

Solution. Let's think about this with casework and just divide by 2^{10} at the end: If 0 pairs of tails, then 1 for this to occur.

If we have 1 pair only, we have 9 spots for the pair.

If we have 2 pairs, we can think of it as 2 groups along with 6 heads that we need to arrange in a line. This gives us 8 choose 2, which is 28.

For 3 pairs, we have 3 groups and 4 heads, so $\binom{7}{3} = 35$. For 4 pairs, we have 4 groups and 2 heads, so 15 ways.

For 5, we have 5 groups, so 1 way.

Our answer is then $(10 + 36 + 28 + 1)/1024 = \boxed{\frac{89}{1024}}$

MEDIUM

Tags: Brainteasers

Topics: Other

Date: May 6, 2026

13. Paired Pumpkins II**Problem**

Dracula has 5 pumpkins. When he pairs any two pumpkins, they weigh 21, 22, $\dots$, 30 pounds. What's the sum of the weights of all the pumpkins?

Solution. Consider each of the pumpkins. Note that it can be paired with one of the other 4 pumpkins, meaning each pumpkin is used a total of 4 times for weighing. This means we simply need to add up the weights and divide by 4 to get $\boxed{\frac{255}{4}}$ pounds as our answer.

HARD

Tags: Probability

Topics: Continuous Random Variables

Date: May 7, 2026

14. Square Ratio**Problem**

Suppose that $X, Y \sim \text{Unif}(0, 1)$ i.i.d. Compute the probability that $\left\lceil \frac{X}{Y} \right\rceil$ is a perfect square. The answer is in the form $q - \frac{\pi^2}{a}$ for a rational q and an integer a . Find aq .

Solution. Note that we have a π^2 in the desired form, and that probably means that it's of interest to use geometric probability. Recall that $\left\lceil \frac{Y}{X} \right\rceil = k$ implies that $k - 1 < \frac{Y}{X} \leq k$. Since we want the region where $k = n^2$, our equation of interest turns into $(n^2 - 1)X < Y \leq n^2X$. Now, n can be any value, and so we must evaluate the area created by all values of n . For a particular value of n , we have that area to be

$$\int_0^1 \left(\frac{1}{n^2 - 1} - \frac{1}{n^2} \right) Y \, dY = \frac{1}{2} \left(\frac{1}{2} \left(\frac{1}{n - 1} - \frac{1}{n + 1} \right) - \frac{1}{n^2} \right)$$

for $n \geq 2$ (we used partial decomposition of fractions at the end). As for $n = 1$, we know that the area is simply $\frac{1}{2}$.

Now, we sum our expression from 2 to infinity. We see that our telescoping series evaluates to $\frac{1}{4} \cdot \frac{3}{2}$ while our other series sums up to $\frac{1}{2} \left(\frac{\pi^2}{6} - 1 \right)$. Our sum is thus,

$$\frac{1}{2} + \frac{3}{8} - \left(\frac{\pi^2}{12} - \frac{1}{2} \right) = \frac{11}{8} - \frac{\pi^2}{12} \rightarrow a = 12, q = \frac{11}{8}$$

$$aq = \boxed{\frac{33}{8}}$$

as desired.

HARD

Tags: Probability

Topics: Expected Value

Date: May 8, 2026

15. Decreasing Uniform Chain

Problem

Let $X_1, X_2, \dots \sim \text{Unif}(0, 1)$ i.i.d. Let N be the first index n where $X_n \neq \min\{X_1, \dots, X_n\}$. Find $\mathbb{E}[X_{N-1}]$, i.e., the smallest value among the first N values selected. The answer will be in the form $a + be$ for integers a and b . Note here that e is Euler's constant. Find $a + b$.

Solution. This problem was very difficult, and I don't think I actually live-solved it :(. But here's the approach AI and I came up with:

Suppose $N = n$. We must first find the probability density function of X_{n-1} , let this be $f_{n-1}(y)$. Suppose the value of X_{n-1} is y . Note that the probability that the past $n - 2$ X_i being greater than X_{n-1} is equal to $(1 - y)^{n-2}$, and for those terms to be in decreasing order, this would make our probability be $\frac{1}{(n-2)!} (1 - y)^{n-2}$ (think about the ordering of the $n - 2$ variables). Finally, we must consider X_n , which has a probability of $1 - y$ of being greater than X_{n-1} .

This gives our probability density function as $f_{n-1}(y) = 1 \cdot \frac{(1-y)^{n-1}}{(n-2)!}$ where the 1 comes from the original PDF of X_{n-1} .

Recall that n can be any integer $n \geq 2$. Thus, our actual probability density function is

$$f(y) = \sum_2^{\infty} \frac{(1-y)^{n-1}}{(n-2)!} = (1-y)e^{1-y}$$

where the last step is noticing that the summation with one of the $(1 - y)$ factors is the Taylor series for e^x .

We now compute the integral to find our answer, which is

$$\mathbb{E}[X_{N-1}] = \int_0^1 y(1-y)e^{1-y} dy = 3 - e$$

from integration by parts. Thus, our answer is $3 - 1 = \boxed{2}$.

HARD

Tags: Probability

Topics: Expected Value

Date: May 9, 2026

16. Collecting Toys II

Problem

Every box of cereal contains one toy from a group of 5 distinct toys, each of which is mutually independent from the others and is equally likely to be within a given box. How many distinct toys can you expect to collect if you bought 7 boxes?

Solution. The key to solving this problem is to assign an indicator for each of the toys $I_1, I_2, \dots, I_5$, where $I_i = 1$ if it does appear in the 7 boxes. We now simply find the expected value of all these indicators. For I_i , the probability that it does appear is $1 - \left(\frac{4}{5}\right)^7$ via

complementary probability. Thus, our answer is simply $5 \left(1 - \left(\frac{4}{5}\right)^7\right)$.

HARD

Tags: Probability

Topics: Expected Value

Date: May 9/10, 2026

17. Close Dice II

Problem

On average, how many rolls of a fair 6-sided die are required to get two consecutive rolls with a difference of at most 1?

Solution. I didn't solve this all the way through because it got too tedious. Anyways, this is a classic Markov Chain problem. Suppose S_i is the expected number of rolls after rolling the number i . By symmetry, we have $S_1 = S_6$, $S_2 = S_5$, $S_3 = S_4$. And the following equations:

$$S_1 = 1 + \frac{1}{6}(S_3 + S_4 + S_5 + S_6) = 1 + \frac{1}{6}(2S_3 + S_2 + S_1)$$

$$S_2 = 1 + \frac{1}{6}(S_4 + S_5 + S_6) = 1 + \frac{1}{6}(S_1 + S_2 + S_3)$$

$$S_3 = 1 + \frac{1}{6}(S_1 + S_5 + S_6) = 1 + \frac{1}{6}(2S_1 + S_2)$$

and solving this system of equation yields $S_1 = \frac{288}{115}$, $S_2 = \frac{246}{115}$, $S_3 = \frac{252}{115}$. Thus, our

answer is $E = 1 + \frac{1}{6}(S_1 + S_2 + S_3 + S_4 + S_5 + S_6) = 1 + \frac{1}{3}(S_1 + S_2 + S_3) = \boxed{\frac{377}{115}}$.

Tags:

Topics:

Date:

18.

Problem

Solution.